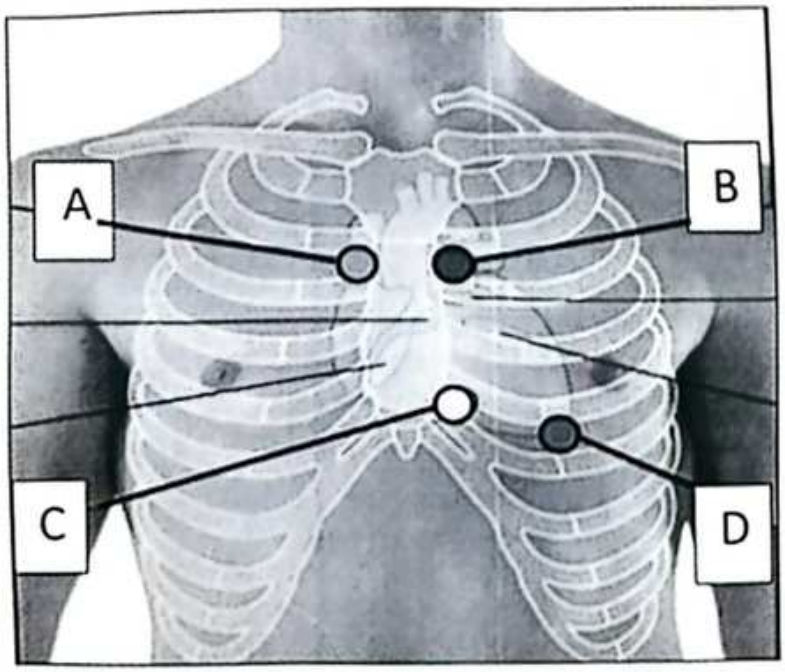


A1

Q1) Which heart sound can be best heard at position D?

a) First heart sound S1

b) Second heart sound S2

c) Third heart sound

d) Fourth heart sound

Q2) Name the valve which produce sound heard at positions B?

a) Aortic valve

b) Pulmonary valve

c) Tricuspid

d) Mitral

Q3) A heart murmur indicates a defect in which one of the following structure?

a) Bundle of His

b) SA node

c) AV node

d) Heart valves

Q4) Which of the following events is associated with the first heart sound?

a) Closing of the aortic valve

b) Beginning of diastole

c) Opening of the A-V valves

d) Closing of the A-V valves

Q5) First heart sound occurs during which of the following period?

a) isometric relaxation

b) isotonic relaxation

c) isovolumetric contraction

d) isovolumetric relaxation

9

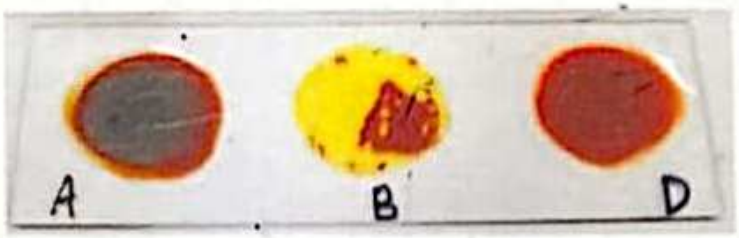


Q1. Identify the contractile protein shown in above figure. **(1 mark)**

Q2. Name the molecule labeled A. Where does it bind on the above structure?

(2 marks)

Q3. Name the structure labeled B and write its function. **(2 marks)**

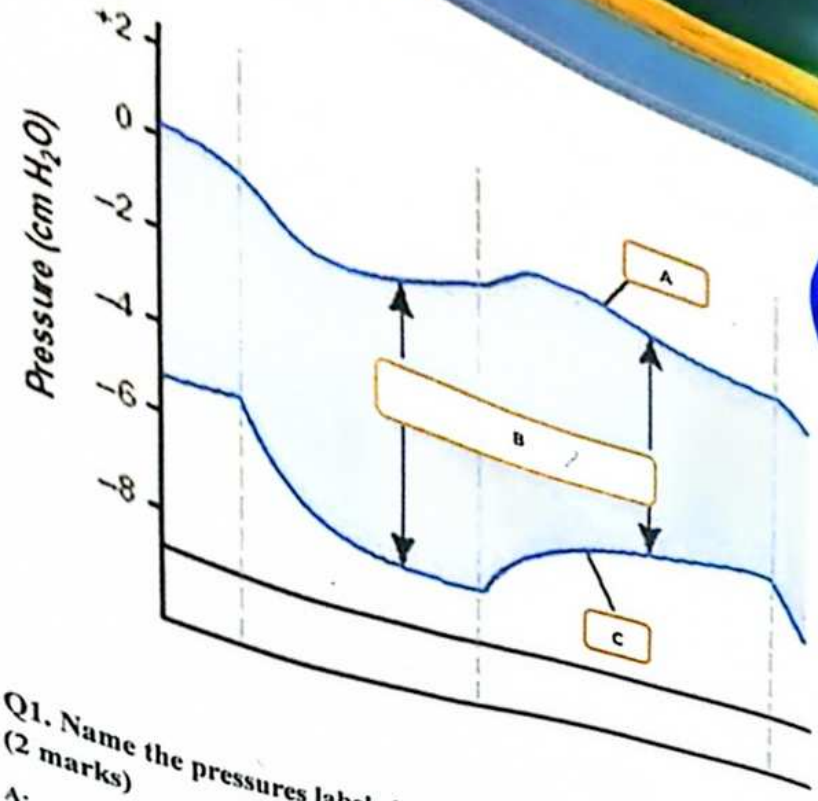


Anti-A antibody Anti-B antibody Anti-D antibody

Q1. Identify the blood group with the help of above slides? (2 marks)

Q2. Which antibody will be present in the plasma of the person with above blood group? (2 mark)

Q3. Which blood group is the universal donor in ABO system? (2 marks)



6

Q1. Name the pressures labeled as A and C in the above graph? (2 marks)

A: _____

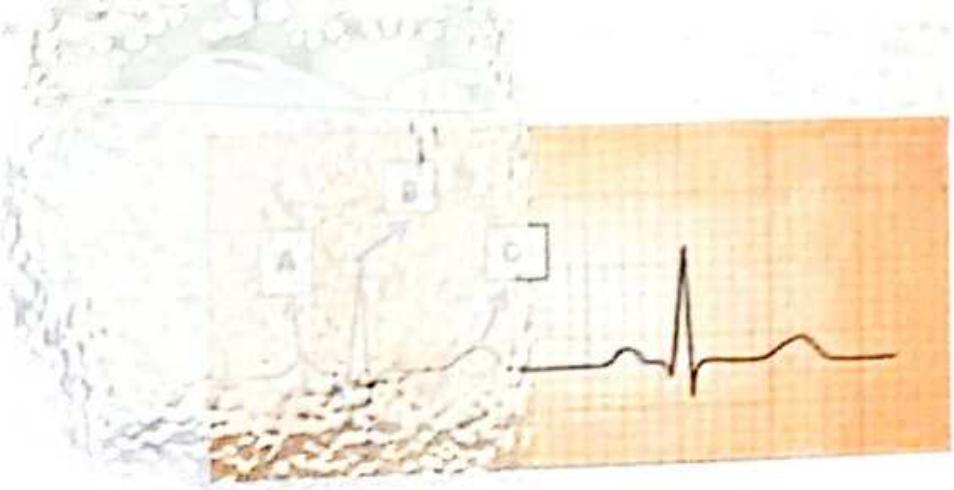
C: _____

Q2. Name the pressure labeled B in the above graph? (1.5 marks)

_____

Q3. What is the value of pressure A during inspiration? (1.5 marks)

_____



7

Q1. Name the wave labeled A and its cause of formation.

(2 marks)

_____

Q2. Name the wave labeled B and its cause of formation.

(2 marks)

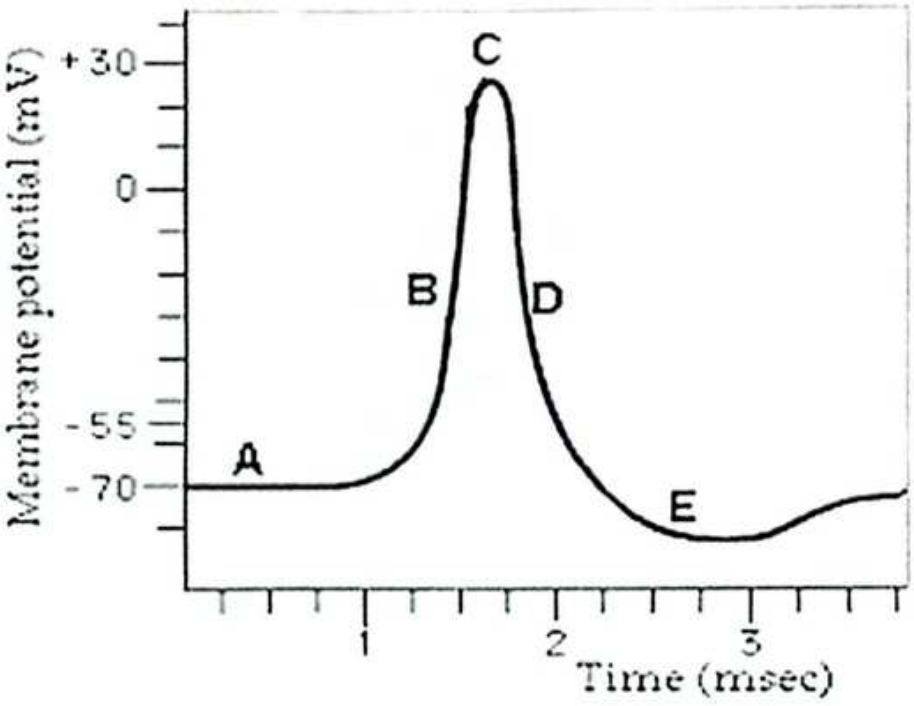
_____

Q3. Name the wave labeled C.

(1 mark)

_____

3



Q1. Identify the stage B in the above diagram. **(1 mark)**

B: _____

Q2. Which channels open during this stage and which ions flow inside cell?

(2 marks)

_____

Q3. Identify the stage D and channels involved in it. **(2 marks)**

D: _____

Q1. In the ABO system, blood group 'O' is characterized by which one of the following criteria?

a) presence of antigen O
b) presence of both antigens A and antigen B
c) absence of both antigens A and antigen B
d) presence of antigen A and absence of antigen B
e) Presence of antigen B and absence of antigen A

Q2. Antiserum is defined as which of the following?

(a) blood serum containing specific antibodies
(b) blood serum containing specific antigens
(c) blood serum containing a mixture of antigens and antibodies
(d) blood serum in which antigens and antibodies are both absent

Q3. Which antibodies are found in the plasma of a person with type A blood?

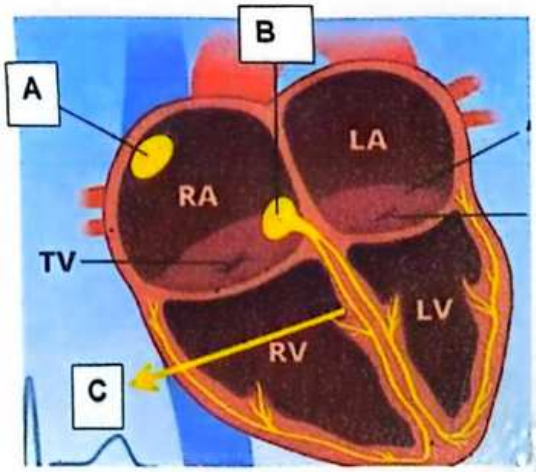
a) anti-A, but not anti B
b) neither anti-A nor anti B
c) both anti-A and anti B
d) anti-B, but not anti-A

Q4. Human blood types in the ABO blood group are identified by which one of the following?

a) clotting factors in plasma
b) microscopic examination of white blood cells
c) antigen-antibody reactions
d) series of enzyme controlled reactions

Q5. Which of the following blood group is referred to a universal recipient?

A
O
AB



(5)

Q1. Identify the structures labeled A and B in the above diagram.

(2 marks)

A: _____

B: _____

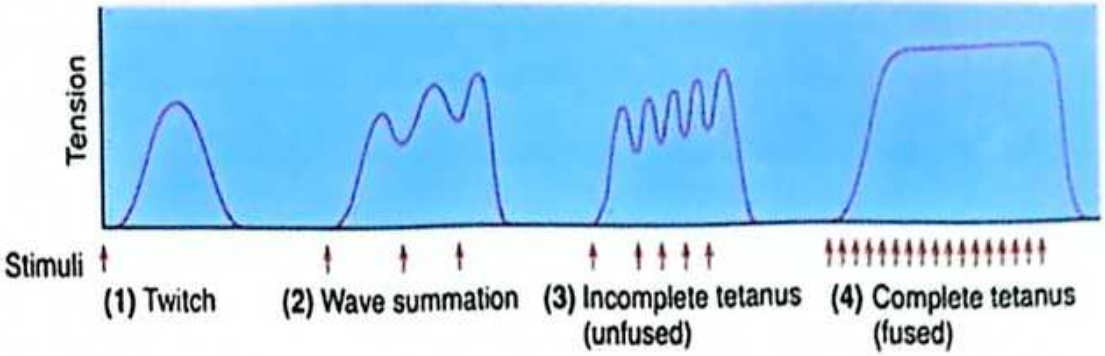
Q2. What is the rate of impulse production by structure labeled A?

(2 marks)

_____

Q3. Name the structure labeled C. **(1 mark)**

_____



Q 1. Which of the following is defined as lack of relaxation between successive stimuli in sustained muscle contraction?

a) Fatigue
b) Tetanus
c) Tonus
d) Spasm

Q2. Which one of the following is the trigger to initiate the contractile process in skeletal muscle?

a) calcium binding to tropomyosin
b) ATP binding to the myosin cross bridge
c) calcium binding to troponin.
d) potassium binding to myosin

Q 3. The fundamental physiological event that occurs when a muscle fiber contracts in response to an action potential from a motor neuron is defined as?

a) Muscle twitch
b) Wave summation
c) Fatigue
d) Tetanus

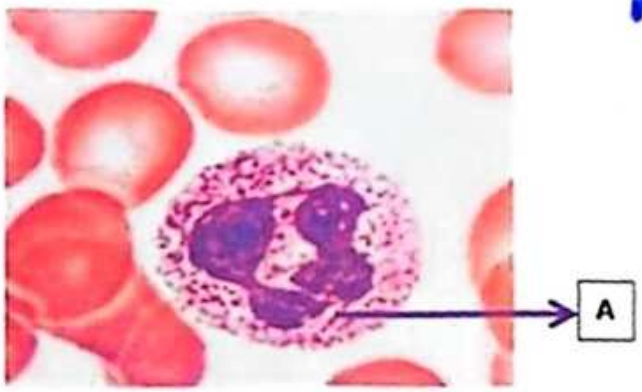
Q 4. Weight lifting can result in a dramatic increase in skeletal muscle mass. This increase in muscle mass is primarily attributable to which of the following?

a) Fusion of sarcomeres between adjacent myofibrils
b) Hypertrophy of individual muscle fibers
c) Increase in skeletal muscle blood supply
d) Increase in the number of motor neurons

Q 5. Which one of the following is the reason for muscle fatigue?

a) Inability of the action potential to spread over the muscle
b) Failure of transmission in the motor nerve
c) Failure of neuro muscular transmission
d) Accumulation of lactic acid

B 8



Q1. Identify the cell labeled A?

a) Lymphocytes
b) Neutrophils
c) Eosinophils
d) Basophils

Q2. Which one of the following is the normal count of this cell in percentage?

a) 2.3%
b) 62.0%
c) 0.4%
d) 5.3%

Q3. What is the primary function of this cell in the immune response?

a) To recognize and bind to antigens
b) To produce antibodies
c) To destroy pathogens through phagocytosis
d) To activate T cells

Q4. What is the process by which these leucocytes are attracted to sites of infection or injury?

a) Phagocytosis
b) Chemotaxis
c) Opsonization
d) Activation

Q5. Neutrophils are capable of producing extracellular traps (NETs). What is the purpose of NETs?

a) To bind to antigens and prevent their spread
b) To create a physical barrier around pathogens
c) To trap and kill bacteria
d) To activate the complement system

B 7

FIGURE 1

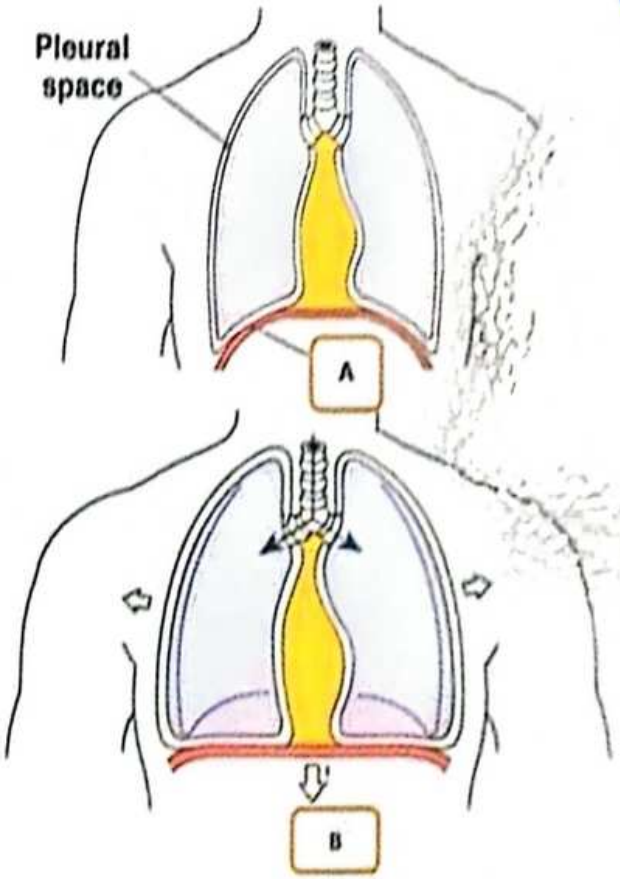
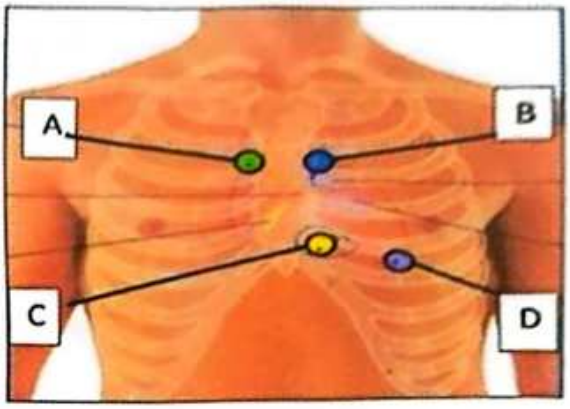


FIGURE 2

Q1. Name the muscle labeled A and write its function during inspiration. (2)

Q2. Which respiratory phase is shown in the diagram B? (1.5)

Q3. How much volume of air is taken in lungs during the phase shown in diagram B? & what this volume is called? (1.5)



B 3

Q1) Which heart sound can be best heard at position D?

a) First heart sound S1

b) Second heart sound S2

c) Third heart sound

d) Fourth heart sound

Q2) Name the valve which produce sound heard at positions B?

a) Aortic valve

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Q3) A heart murmur indicates a defect in which one of the following structure?

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d) Closing of the A-V valves

Q5) First heart sound occurs during which of the following period?

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b) isotonic relaxation

c) isovolumetric contraction

d) isovolumetric relaxation

B-12



Q 1- Identify the name of the instrument?

a) Hematology analyzer
b) Hemoglobinometer
c) Sahli's haemometer
d) Westergens tube

Q 2- Which one of the following terms defines the quantity or amount of hemoglobin present in one RBC?

a) Mean Corpuscular Volume
b) Mean Corpuscular Hemoglobin
c) Mean Corpuscular Hemoglobin Concentration
d) Hematocrit

Q 3 - Fetal Hb consist of

a) 3 alpha chains+ 2 beta chains
b) 2 alpha chains+ 2 delta chains
c) 2 alpha chains +2 beta chains
d) 2 alpha chains+ 2 gama chains

Q 4- Which of the following is the oxygen binding site of the hemoglobin?

a) N-terminal of the beta subunit

b) Carboxyterminal of both alpha and beta subunits

c) Ferric ion (Fe+3) of the heme molecule

d) Ferrous ion (Fe+2) of the heme molecule

Q 5- Sickle cell disease is an inherited blood disorder of hemoglobin synthesis, caused by which of the following mutation?

a) In the sixth position of the alpha chain (glutamic acid is replaced by isoleucine)
b) In the sixth position of the beta chain (glutamic acid
c) In the sixth (6th) position of the α-chain (Glutamic acid is replaced by Valine)
d) In the sixth (6th) position of the β-chain (Glutamic acid is replaced by Valine)

B 19



Q1. What is the name of the instrument used to measure Blood pressure?

a) Manometer
b) Sphygmomanometer
c) Barometer
d) Thermometer

Q2. During taking BP at upper arm, chest piece of stethoscope is placed in relation to which of the following artery?

a) Axillary
b) Brachial
c) Popliteal
d) Radial

Q3. Which one of the following is the normal Blood pressure value?

a) 130/90 mm Hg
b) 130/80 mm Hg
c) 120/80 mm Hg
d) 110/90 mmHg

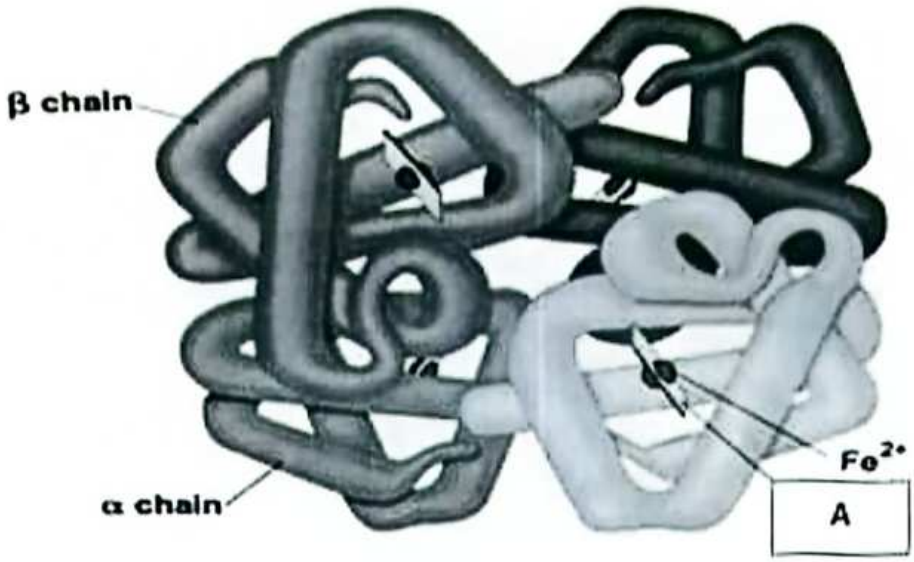
Q4. Which one of the following is defined as Pulse Pressure?

a) Pressure in pulmonary veins
b) Difference between systolic and diastolic pressure
c) Average pressure produced during the cardiac cycle
d) Maximum pressure produced during the cardiac cycle

Q5. If Blood pressure is 160/100, what will be the mean arterial blood pressure?

a) 130
b) 125
c) 110
d) 120

B6



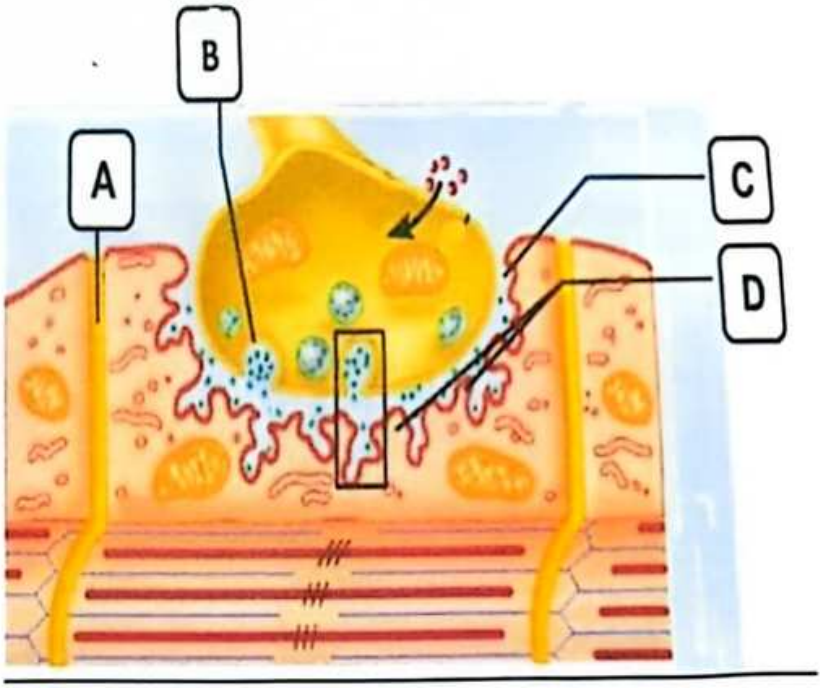
Q1. Identify the structure marked as A in the given diagram?
a) Alpha chain
b) Beta chain
c) Heme group
d) Porphyrin

Q2. Adult hemoglobin (HbA) is a tetrameter consisting of which one of the following chains?
a) α2β2
b) α2δ2
c) α2γ2
d) β4

Q3. How many heme groups are present in a single hemoglobin molecule?
a) 1
b) 2
c) 4
d) 8

Q4. What is the role of hemoglobin in carbon dioxide transport?
a) It carries carbon dioxide dissolved in plasma.
b) It breaks down carbon dioxide into its constituent elements.
c) It converts carbon dioxide into a soluble form for transportation.
d) It binds to carbon dioxide and releases it in the lungs.

Q5. The 1st step of Heme synthesis takes place in which of the following organelles?
a) Mitochondria
b) Golgi Apparatus
c) Rough Endoplasmic Reticulum
d) Peroxisomes



Q1. Identify the structure shown in the above figure. **(1 mark)**

_____

Q2. Name the structures labeled A and B in the above figure. **(2 marks)**

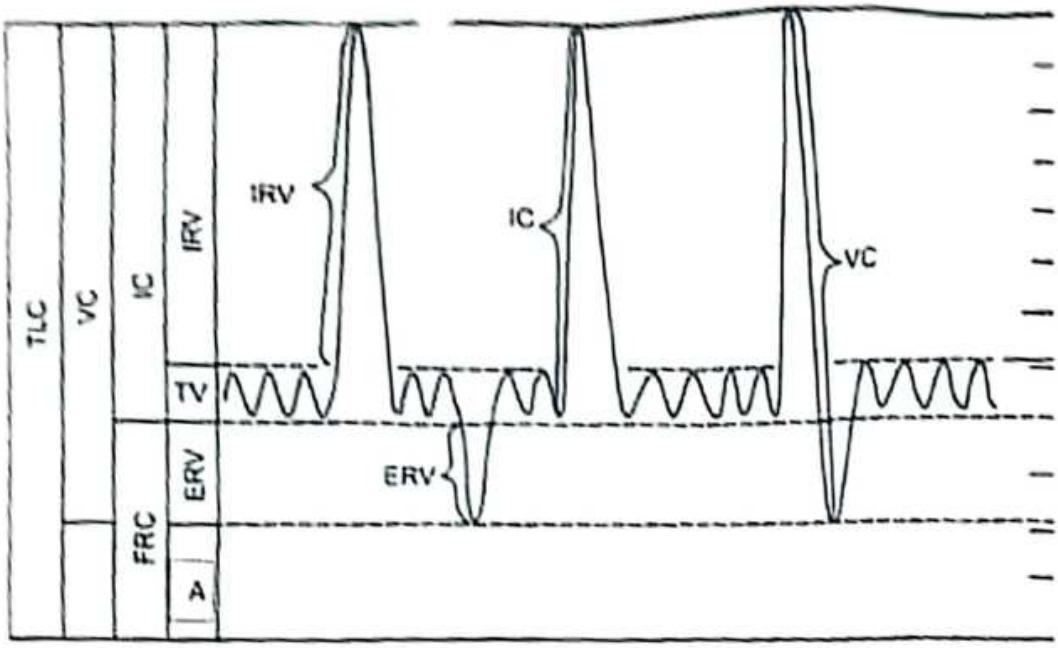
A: _____

B: _____

Q3. Name the neurotransmitter released by structure labeled B. **(2 marks)**

_____

B-15



Q1) Identify the given diagram?

a) Electroencephalogram
b) Electromyogram
c) Spirogram
d) Electrocardiogram

Q2) Identify the structure marked as A?

a) Residual Volume
b) Tidal Volume
c) Vital Capacity
d) Total Lung Capacity

Q3) Which of the following lung volumes or capacities can be measured by spirometry?

a) Functional residual capacity
b) Residual Volume
c) Vital Capacity
d) Total lung capacity

Q4) An average adult male has a vital capacity of which one of the following volume?

a) 3 liters
b) 4 liters
c) 5 liters
d) 6 liters

Q5) Which one of the following is the normal value of Functional Residual Capacity?

a) 3500 ml
b) 2300 ml
c) 4600 ml
d) 5800 ml

B(1)



Q1. Name the test being performed in above picture? (1.5 Mark)

_____

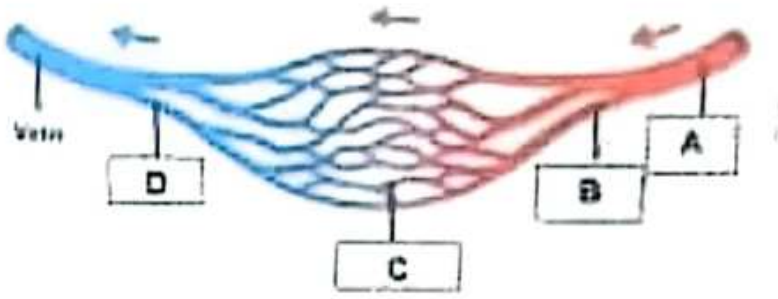
Q2. What is the normal range of this test? (1.5 Mark)

_____

Q3. Name any two condition in which this test is prolonged. (2 marks)

_____

B6



Q1. Name the structures labeled A and B. (2 marks)

A:

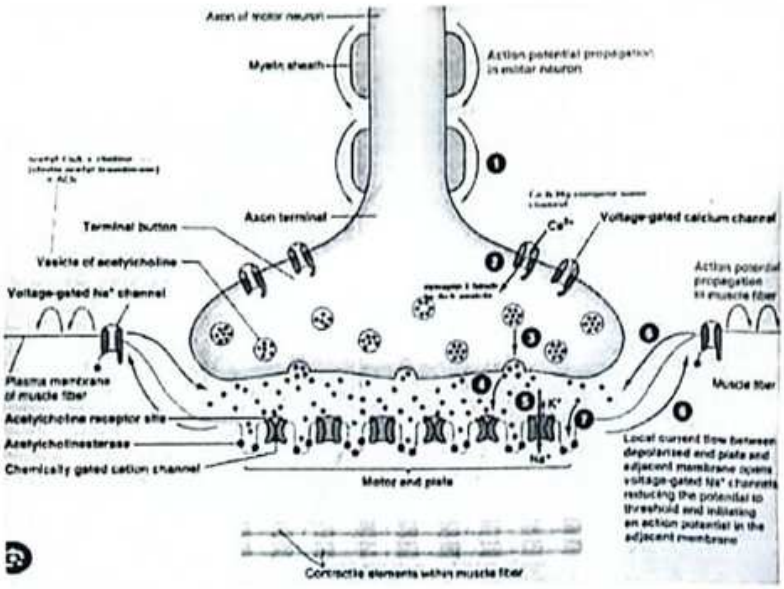
B:

Q2. Write one function of structure labeled B. (1 mark)

Q3. Name the structure labeled C and write its one function. (2 marks)

C:

B-14



Q1. What is the space between the axon terminal & the muscle cell membrane called?

a) Motor end-plate
b) Myelin sheath
c) Synaptic cleft
d) Motor neuron

Q2. Where does the stimulation of muscle fibers by a motor neuron take place?

(a) myofibril
(b) transverse tubules
(c) sarcoplasmic reticulum
(d) neuromuscular junction

Q3 An action potential in the motor end plate rapidly spreads to the central portions of the muscle cells by means of which one of the following factors?

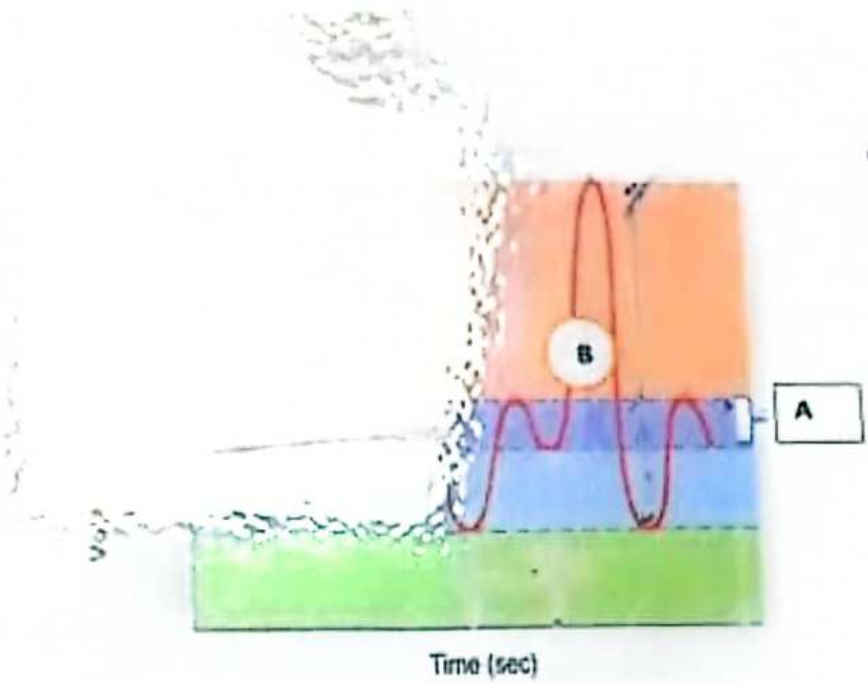
a) Z -Lines
b) Sarcoplasmic reticulum
c) Pore in the plasma membrane
d) Transverse tubules

Q4. Select the best option regarding the neuromuscular junction?

a) The muscle membrane possesses muscarinic Receptors
b) There is a one-to-one transmission of excitatory impulses from the motor neuron to the muscle fibres it innervates.
c) The motor nerve endings secrete noradrenaline
d) The typical summed end plate potential (EPP) is usually 10 times the potential necessary to trigger an action potential.

Q5. 42-year-old man with myasthenia gravis noticed increased muscle strength when he is treated with an acetylcholinesterase (AChE) inhibitor. The basis for his improvement is increased levels of which of the following?

a) Amount of acetylcholine (ACh) released from motor nerves
b) Levels of ACh at the muscle end plates
c) Number of ACh receptors on the muscle end plates
d) Amount of norepinephrine released from



1. Name the lung volume / capacity shown as A in the above graph and what its normal value? **(2 marks)**

_____

2. Name the lung volume / capacity labeled B. What is its normal value?

(2 marks)

_____

3. Name any one the volume/capacity that cannot be measured by the irometry? **(1mark)**

_____